Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2020	ENGI3391-WE01

Title: MEng and BEng Engineering

Control and Signal Processing 3

Time Allowed:	3 hours	
Additional Material provided:	Signal Processing and Control formula sheets at the end of the paper.	
Materials Permitted:	None.	
Calculators Permitted:		Models Permitted: Those from the Casio fx-83 and fx-85 series.
Visiting Students may use dictionaries: YES		

Instructions to Candidates:	Answer ALL questions. Use a separate answer book for each section. All relevant workings must be shown.

Revision:	

Section A

Question 1

(a) Find the impulse response of the low pass filter LPF in Figure Q.1.a with a cut-off frequency equal to ω_n rad/sec.

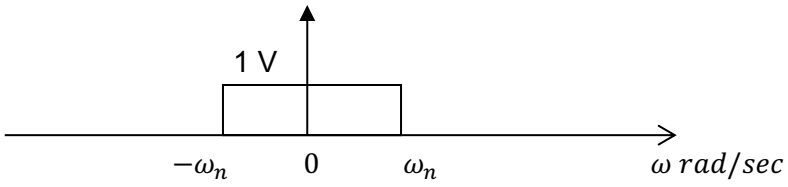


Figure Q.1.a

[40%]

(b) Given the response of the filter in part (a), deduce the impulse response of the band-stop filter in Figure Q.1.b.

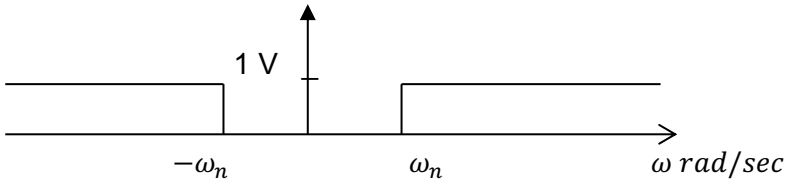


Figure Q.1.b

[25%]

(c) Discuss four advantages to digital implementation of signal processing.

[10%]

(d) Explain what is meant by a time invariant system.

[10%]

(e) Distinguish between the impulse response and the unit step response of a linear time invariant system and give the relationship between the two responses.

[15%]

continued

Question 2

(a) Outline the necessary steps to implement the convolution of *x(t)* with *y(t)*.

[10%]

(b) Determine whether each system possesses the indicated properties:

(i) $y[n] = n^2\, x[n]$ time-invariance

(ii) $h[n] = (0.8)^n\, u[n+1]$ causality and stability

where *y[n]* is the output of the system due to an input *x[n], h[n]* is the impulse response of the system, and *u[n]* is the unit step function.

[25%]

(c) For the system shown in Figure Q.2, where $\delta(t)$ is the impulse function, write an expression for the output *y(t)* in terms of *x(t)* and find the magnitude and phase of the frequency response.

$$H(\omega) = \frac{Y(\omega)}{X(\omega)}$$

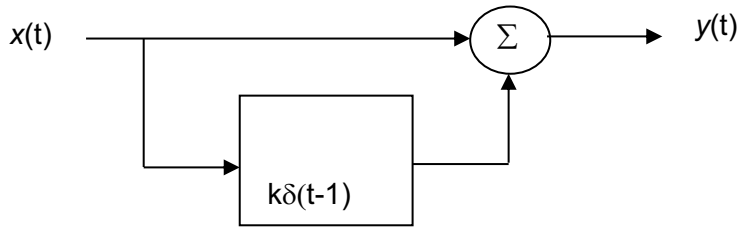


Figure Q.2

[45%]

(d) A system has two natural frequencies at ω_1 =18π rad/sec and ω_2 =22π rad/sec. Assuming that the initial conditions are such that the response of the system is given by:

$$\theta_1(t) = cos(\omega_1 t) + cos(\omega_2 t)$$

Using trigonometric identities find the frequency response of the envelope and the centre frequency of the system's response in the time domain.

[20%]

continued

Section B

Question 3

A plant process is described by the following transfer function:

$$G(s) = \frac{1.5}{(s+2)(s+4)}$$

and is placed in a closed-loop control system as shown in Figure Q3.1.

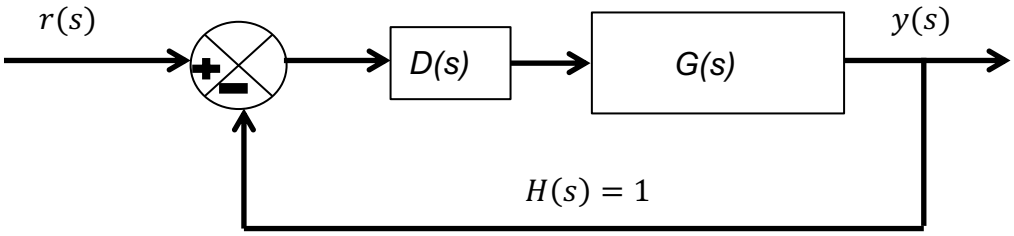


Figure Q3.1: A closed-loop control system with negative unit feedback

(i) What System Type is the plant *G(s)* and what effect will this have on the steady state error of the system if *D(s)* is configured as a gain controller?

[10%]

(ii) Use the provided graph paper to sketch the root locus of this closed-loop system with necessary steps of the root locus plotting. If the compensator *D(s)* is configured as a gain controller to satisfy a performance specification of damping ratio $\zeta = 0.5$ and natural frequency $\omega_n = 6$ rad/s, mark the position of the required closed-loop pole to meet the performance specification.

[20%]

(iii) If the compensator *D(s)* is configured as a gain controller, use the magnitude criteria to determine the gain required to meet the performance specification in (ii).

[20%]

(iv) If the compensator *D(s)* is configured as an ideal phase-lead compensator, i.e., *D(s)= k(s+z)*, to satisfy a performance specification of damping ratio $\zeta = 0.8$ and natural frequency $\omega_n = 10$ rad/s, use both angle and magnitude criteria to design this compensator.

[30%]

(v) Discretise this designed compensator in (iv) by using the backward difference algorithm, i.e., $z = \frac{1}{1-sT}$ (assuming the sampling interval *T=0.1* second) that is suitable for being implemented in a microprocessor.

[20%]

Question 4

The stability of linear time invariant systems can be analysed in both the time domain and the frequency domain.

(i) Explain why it may be desirable to use frequency response methods to determine the stability of a closed-loop system rather than the time domain methods.

[10%]

(ii) State the difference between Bode Diagram and Nyquist Diagram, and comment on the advantages and disadvantages of these two methods.

[20%]

(iii) The open loop system is of the form:

$$GH(s) = \frac{K}{s(s+2)(s+4)}$$

When *K=40*, the open loop frequency response data is given in Table Q4 below. Present this system's frequency response using a Bode Diagram.

[30%]

ω (rad/s)	Gain (dB)	Phase (degree)
0.5	19.5	-110
0.6	18.0	-115
0.7	16.5	-120
0.8	15.0	-123
1.0	12.5	-130
1.5	8.0	-148
2.0	4.0	-160
3.0	-2.5	-183
4.0	-8.0	-199

Table Q4

(iv) Using your Bode Diagram, calculate the gain and phase margin and comment on the relative stability of the system.

[20%]

(v) Calculate the value of *K* that would provide a phase margin of 65 degrees and comment on the relative stability of the system with this new value of *K*.

[20%]

continued

L3 Signal Processing Formula Sheet

Energy of a signal

$$E = \int_{-\infty}^{\infty} |f(t)|^2 dt$$

Power of a signal

$$P_{av} = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} |f(t)|^2 dt$$

Fourier series

$$x(t) = a_o + \sum_{k=1}^{\infty} \left[a_k \cos k\omega_o t + b_k \sin k\omega_o t\right]$$

where

$$a_o = \frac{1}{T}\int_T x(t)dt = dc\ term$$

$$a_k = \frac{2}{T}\int_T x(t)\cos k\omega_o t dt \qquad b_k = \frac{2}{T}\int_T x(t)\sin k\omega_o t dt$$

Alternative Fourier series

Trigonometric form:

$$x(t) = a_o + \sum_{k=1}^{\infty} \sqrt{a_k^2 + b_k^2}\ \cos(k\omega_o t + \phi_k)$$

where

$$\phi_k = -\tan^{-1}\frac{b_k}{a_k}$$

Complex exponential form

$$x(t) = \sum_{k=-\infty}^{\infty} d_k\, e^{jk\omega_o t}$$

where $d_k = \frac{1}{T}\int_T x(t).e^{-jk\omega_o t} dt$

and $d_k = \frac{1}{2}(a_k - jb_k)\ for\ k > 0\ and\ d_k = \frac{1}{2}(a_k + jb_k)\ for\ k < 0$

continued

Fourier transform

$$\mathcal{F}\{x(t)\} = X(\omega)$$
$$= \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$$

Alternatively, it can be written in terms of the frequency in cycles per seconds or Hz as

$$X(f) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft}dt$$

Inverse Fourier transform

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)e^{j\omega t}d\omega$$

or

$$x(t) = \int_{-\infty}^{\infty} X(f)e^{j2\pi ft}df$$

Fourier transform properties

1. Linearity

$$x_1(\mathrm{t}) \leftrightarrow X_1(\omega), \text{and } x_2(\mathrm{t}) \leftrightarrow X_2(\omega) \text{ then}$$
$$\mathrm{a}x_1(\mathrm{t}) + bx_2(t) \leftrightarrow aX_1(\omega) + bX_2(\omega)$$

2. Differentiation

$$y(t) = \frac{d^n x(t)}{dt^n} \leftrightarrow (j\omega)^n X(\omega)$$

3. Integration

$$y(t) = \int_{-\infty}^{t} x(\tau)d\tau \leftrightarrow \frac{1}{j\omega}X(\omega) + \pi X(0)\delta(\omega)$$

4. Time scaling

$$x(b\,\mathrm{t}) \leftrightarrow \frac{1}{|b|}X\left(\frac{\omega}{b}\right)$$

5. Time reversal

$$x(-\mathrm{t}) \leftrightarrow X(-\omega) = X^*(\omega)$$

6. Time delay

$$y(t) = x(\mathrm{t}\text{-}\mathrm{T}) \leftrightarrow X(\omega)\,\mathrm{e}^{-\mathrm{j}\omega\mathrm{T}}$$

7. Multiplication in time

$$x_1(\mathrm{t}).x_2(\mathrm{t}) \leftrightarrow \frac{X_1(\omega) * X_2(\omega)}{2\pi}$$

continued

8. Multiplication in frequency (convolution in time)

$$X_1(\omega)X_2(\omega) \leftrightarrow x_1(t) * x_2(t)$$

9. Modulation property

$$x(t)\cos(\omega_o t) \leftrightarrow \frac{1}{2}X(\omega - \omega_o) + \frac{1}{2}X(\omega + \omega_o)$$

10. Duality property

If $x(t) \leftrightarrow X(\omega)$

Then $X(t) \leftrightarrow 2\pi x(-\omega)$

Discrete Fourier Transform

Discrete Fourier Series

$$d_k = X[k] = \frac{1}{N}\sum_{n=0}^{N-1} x[n]e^{\frac{-j2\pi kn}{N}}$$

Inverse Discrete Fourier Series

IDFS: $$x[n] = \sum_{k=0}^{N-1} d_k\, e^{\frac{j2\pi kn}{N}}$$

Euler's identities

$$A\sin(\omega_o t) = \frac{A}{j2}\left(e^{j\omega_o t} - e^{-j\omega_o t}\right)$$

$$A\cos(\omega_o t) = \frac{A}{2}\left(e^{j\omega_o t} + e^{-j\omega_o t}\right)$$

$$A\cos(\omega_o t + \phi_o) = \frac{A}{2}\left(e^{j\omega_o t}e^{j\phi_o} + e^{-j\omega_o t}e^{-j\phi_o}\right)$$

$e^{\pm j\beta} = \cos\beta \pm j\sin\beta$

Impulse function in continuous time domain

$\delta(t) = 0\ for\ t \neq 0\ and\ infinite\ at\ t = 0$

$$\int_{t=-\infty}^{\infty} \delta(t) = 1$$

Impulse function in discrete time domain

$\delta[n] = 1\ n = 0\ and\ 0\ elsewhere$

Unit step function in continuous time domain

continued

$u(t) = 1\ for\ t \geq 0\ and\ 0\ t < 0$

Unit step function in discrete time domain

$u[n] = 1\ for\ n \geq 0\ and\ 0\ for\ n < 0$

Convolution integral

$$y(t) = x_1(t) * x_2(t) = \int_{-\infty}^{\infty} x_1(\tau)x_2(t-\tau)d\tau = \int_{-\infty}^{\infty} x_1(t-\tau)x_2(\tau)d\tau$$

Convolution sum

$$y[n] = \sum_{k=-\infty}^{\infty} x_1[k]x_2[n-k] = \sum_{k=-\infty}^{\infty} x_1[n-k]x_2[k]$$

Convolution with an impulse

$x(t) * \delta(t - t_o) = x(t - t_o)$

Correlation functions

Fourier transform of an impulse

$\delta(t) \leftrightarrow 1$

Inverse Fourier transform of an impulse

$\delta(\omega) \leftrightarrow 1/2\pi$

Roots of a second order polynomial

$ax^2 + bx + c = 0$

are given by

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Trigonometric identities

cos θ = cos(-θ) = sin(π/2 - θ)

sin θ = - sin(-θ) = cos(π/2 - θ)

tan θ = - tan(-θ) = cot(π/2 - θ)

sin 2θ = 2 cos θ sin θ

cos 2θ = $\cos^2\theta$ - $\sin^2\theta$ = 2 $\cos^2$ θ- 1 = 1 - 2 $\sin^2\theta$

tan 2θ = 2 tan θ/ (1 - tan2θ)

continued

$\sin\alpha.\sin\beta = \frac{1}{2}\{(\cos(\alpha-\beta) - cos(\alpha+\beta)\}$

$\cos\alpha.\cos\beta = \frac{1}{2}\{(\cos(\alpha-\beta) + cos(\alpha+\beta)\}$

$\sin\alpha.\cos\beta = \frac{1}{2}\{(\sin(\alpha-\beta) + sin(\alpha+\beta)\}$

$\sin\theta/2 = \pm\sqrt{[(1 - \cos\theta)/2]}$

$\cos\theta/2 = \pm\sqrt{[(1 + \cos\theta)/2]}$

$\tan\theta/2 = \sin\theta/(1 + \cos\theta)$

$\cos^2\theta + \sin^2\theta = 1$

$\sec^2\theta - \tan^2\theta = 1$

$\text{cosec}^2\theta - \cot^2\theta = 1$

$\sin^2\theta = ½(1 - \cos 2\theta)$

$\cos^2\theta = ½(1 + \cos 2\theta)$

$\tan^2\theta = (1 - \cos 2\theta)/(1 + \cos 2\theta)$

$\sin(A + B) = \sin A\cos B + \cos A\sin B$

$\sin(A - B) = \sin A\cos B - \cos A\sin B$

$\cos(A + B) = \cos A\cos B - \sin A\sin B$

$\cos(A - B) = \cos A\cos B + \sin A\sin B$

$\tan(A + B) = (\tan A + \tan B)/(1 - \tan A\tan B)$

$\tan(A - B) = (\tan A - \tan B)/(1 + \tan A\tan B)$

$\sin A + \sin B = 2\sin ½(A + B)\cos ½(A - B)$

$\sin A - \sin B = 2\cos ½(A + B)\sin ½(A - B)$

$\cos A + \cos B = 2\cos ½(A + B)\cos ½(A - B)$

$\cos A - \cos B = -2\sin ½(A + B)\sin ½(A - B)$

$\tan A + \tan B = \sin(A + B)/(\cos A\cos B)$

$\tan A - \tan B = \sin(A - B)/(\cos A\cos B)$

$\sin2A + \sin2B = 1 - \cos(A + B)\cos(A - B)$

$\sin2A - \sin2B = \sin(A + B)\sin(A - B)$

$\cos2A + \sin2B = 1 - \sin(A + B)\sin(A - B)$

continued

cos2A - sin2B = cos(A + B)cos(A - B)

cos2A + cos2B = 1 + cos(A + B)cos(A - B)

cos2A - cos2B = - sin(A + B)sin(A - B)

For a triangle with sides a, b, c, and angles A, B, C opposite sides a, b and c respectively, the following relations hold.

a2 = b2 + c2 - 2 b c cos A

a/sin A = b/sin B = c/sin C.

(a - b)/(a + b) = tan ½(A - B)/ tan ½(A + B)

L3 Control Formula Sheet

Standard second-order transfer function

$$\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Closed-loop poles of under-damped second-order system

$$s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$

The overshoot percentage (%) of the standard second-order system

$$M_o = 100\exp\left(\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}\right)$$

The settling time for a second-order system can be defined as

$$T_s = \frac{\text{Number of time constants}}{\zeta\omega_n}$$

The relation of number of time constants and settling error

Number of time constants	$e^{-\frac{t}{\tau}}$	Settling Error (%)
1	0.368	36.8
2	0.135	13.5
3	0.05	5
4	0.018	1.8
5	0.007	0.7

Defining $u(t)$ as the controller output and $e(t)$ as the error, the standard form of the PID algorithm is:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau)d\tau + K_d \frac{d}{dt}e(t)$$

Laplace form of the PID controller

$$G(s) = K_p + \frac{K_i}{s} + K_d s$$

7 Steps for Root Locus Sketching:

1). Write the characteristic function so that *K* is a multiplier *1+KG(s)=0.*

2). Factor *G(s)* in terms of *n* poles and *m* zeros:

continued

$$G(s) = \frac{\prod_{i=1}^{m}(s - z_i)}{\prod_{j=1}^{n}(s - p_j)}$$

3). Mark poles and zeros on the *s*-plane.

4). Determine the number of separate loci (lines)

5). Locate the segments of the real axis that are root loci

6). The remaining loci proceed to the zeros at infinity along asymptotes centered at σ and with angle Φ

$$\sigma = \frac{\sum_{i=1}^{n} p_i - \sum_{j=1}^{m} z_j}{n - m}$$

$$\phi = \frac{(2q+1)}{n-m}\pi,$$

where $q = 0,1,\ldots,(n-m-1)$

7). Plot remaining root locus.

Magnitude Criteria:

$$K = \frac{\prod_{i=1}^{n}|s - p_i|}{\prod_{j=1}^{m}|s - z_j|}$$

Angle Criteria:

$$\sum_{i=1}^{n}\phi_{p_i} - \sum_{j=1}^{n}\phi_{z_i} = (2q+1)\pi$$

END